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Credit - Drishti Ias

Persistence of Stubble Burning in North India

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For Prelims: Stubble burning, Crop Residue Management (CRM) , Minimum Support Price , The Commission for Air Quality Management , global warming .

For Mains: Causes and consequences of stubble burning, agricultural incentives, air pollution, groundwater depletion, crop diversification and sustainable residue management.

Source: TH

Why in News?

The recurring issue of **stubble burning** in North India, particularly in **Punjab and Haryana** , has renewed concerns over **deteriorating air quality** in Delhi-NCR, highlighting the need to address the **economic, technological and policy factors** driving the practice.

Summary

- Stubble burning persists because of the paddy-wheat cropping cycle, groundwater conservation restrictions, assured procurement and the high cost of alternative residue-management methods.
- Sustainable reduction requires crop diversification, stronger incentives for residue utilisation, affordable technology and improved satellite-based monitoring rather than excessive dependence on penalties.

What is Stubble Burning?

- **About:** Stubble burning refers to the practice of **setting fire to leftover crop residues ,** particularly **straw and stalks,** after harvesting crops such as **paddy and wheat.**
  - It is commonly practised to **clear paddy residues** before wheat sowing, typically between **October and November** .
  - The practice is particularly prevalent in **Punjab, Haryana and parts of Uttar Pradesh.**

Reasons for Stubble Burning

- **MSP-Driven Rice-Wheat Cropping Cycle:** Assured procurement of rice and wheat at **Minimum Support Price (MSP) ,** along with **subsidised electricity,** encourages intensive **rice-wheat cultivation** , discouraging **crop diversification** and generating large quantities of paddy residue.
- **Narrow Sowing Window:** The **Punjab Preservation of Sub-Soil Water Act, 2009** , which restricts early paddy sowing and transplantation to conserve groundwater, has shifted harvesting towards late October and early November.
  - This leaves farmers a **narrow 10–15-day window** before wheat sowing. Long-duration paddy varieties such as **PUSA-44 ,** which take approximately **150–160 days to mature** , further intensify this pressure.
- **Mechanised Harvesting:** Widespread use of combine harvesters leaves substantial standing stubble and loose straw, **increasing residue management costs** and encouraging farmers to **burn residues for rapid field preparation.**
- **Cost-Effectiveness:** Burning is **cheaper and faster** than alternative residue management methods, such as incorporating stubble into the soil or using machinery like **balers and shredders,** which involve additional costs and labour.
- **Weed Management:** Burning can temporarily destroy surface weeds, weed seeds and certain pests, although it is **not a sustainable method** of weed control.
- **Limited Residue Management Options:** Inadequate access to affordable machinery, infrastructure and alternatives such as **composting and bioenergy production** makes burning a convenient option for many farmers.

Impacts

- **Air Pollution:** Stubble burning releases pollutants such as **PM10, PM2.5, nitrogen oxides (NOx), methane (CH_4) , carbon monoxide (CO)** and **volatile organic compounds (VOCs).**
  - These emissions contribute to **smog and deteriorating air quality,** particularly in Delhi-NCR, increasing the risk of respiratory illnesses such as asthma, bronchitis and **Chronic Obstructive Pulmonary Disease (COPD) .**
- **Greenhouse Gas Emissions:** The practice releases **greenhouse gases , contributing to global warming and climate change .**
- **Soil Fertility Loss:** Burning causes the **loss of soil organic matter** and essential nutrients, particularly **nitrogen and sulphur,** while damaging beneficial soil microorganisms.
  - The heat generated can also reduce surface soil moisture, degrading soil health and potentially increasing dependence on chemical fertilisers.

What are the Challenges in Controlling Stubble Burning in India?

- **Technology and Infrastructure Gaps:** Combine harvesters leave substantial standing stubble and loose straw, requiring specialised **Crop Residue Management (CRM) machinery** for timely field preparation.
  - **Custom Hiring Centres (CHCs)** often face shortages and delays in machinery availability during peak harvesting periods, limiting farmers' access to affordable residue management solutions.
  - The **high silica content of paddy straw** complicates industrial processing, while inadequate biomass collection, storage and transportation infrastructure restricts its commercial utilisation.

- **Economic and Funding Limitations:** Despite government subsidies, **high machinery costs, expensive rental services and seasonal utilisation** make alternatives to burning financially unattractive, particularly for small and marginal farmers.
    - The absence of **assured remunerative prices for paddy straw** , coupled with high collection and transportation costs, discourages its use in bioenergy and other industries.
- **Crop Diversification Challenges: MSP-backed procurement of rice and wheat** , along with subsidised electricity, discourages farmers from shifting towards alternative crops, perpetuating the rice-wheat cycle and associated residue management challenges.
- **Policy and Enforcement Challenges:** Inconsistent enforcement, concerns over penalties and differences in state-level incentives complicate the implementation of sustainable residue management practices.
    - Greater reliance on **punitive measures without adequate economic incentives** may discourage farmer cooperation and the adoption of alternatives.
- **Monitoring and Measurement Gaps:** Satellite-based fire monitoring using **MODIS and VIIRS** has limited observation windows, potentially missing fires occurring outside satellite overpass times.
    - Consequently, reductions in reported fire incidents may not accurately reflect changes in the **actual area burned** , complicating the assessment of policy effectiveness.
- **Limited Awareness and Training:** Inadequate access to practical training, technical guidance and awareness programmes on **in-situ and ex-situ residue management** restricts the effective adoption of sustainable alternatives, particularly among small farmers.

India's Initiatives to Tackle Stubble Burning

- **Framework by the Commission for Air Quality Management (CAQM):**
  - **In-Situ Crop Residue Management:** Promotes managing crop residues within fields using **Crop Residue Management (CRM) machinery** , Custom Hiring Centres (CHCs) and bio-decomposers.
  - **Ex-Situ Crop Residue Utilisation:** Encourages alternative uses of paddy straw, including **biomass power generation , co-firing in thermal power plants, second-generation (2G) ethanol, compressed biogas (CBG) production and packaging materials** .
  - **Ban and Enforcement:** Enforces the **prohibition of stubble burning** through satellite monitoring, field inspections and environmental compensation for violations.
- **Financial Support:** The **Crop Residue Management (CRM) Scheme** , implemented since 2018–19, provides **50% financial assistance** to individual farmers for purchasing eligible CRM machinery and **80% assistance** for eligible Custom Hiring Centre projects, subject to prescribed limits.
- **Technological Interventions:**
  - **Happy Seeder:** A tractor-mounted machine that **sows wheat directly into fields containing paddy residues** , reducing the need for burning while retaining straw as soil mulch.
  - **Pusa Decomposer:** A **microbial formulation** that accelerates the decomposition of crop residues into organic matter, improving soil health.
  - **Pelletisation:** Converts **crop residues into biomass pellets** for energy generation, reducing open-field burning and creating opportunities for additional farm income.
  - **Biochar Production:** Converts crop residues into **biochar through pyrolysis** , potentially improving soil fertility, water retention and carbon sequestration.
- **Crop Diversification:** Promotes **short-duration paddy varieties, Direct Seeding of Rice (DSR) and diversification towards less water-intensive crops** to reduce residue generation and ease pressure on the rice-wheat cropping cycle.
- **Local Initiatives:**
  - **Haryana:** Provides financial incentives for **crop residue management, direct seeding of rice and diversification away from paddy** .
    - In 2026–27, the state increased incentives to **₹1,200 per acre for crop residue management, ₹4,500 per acre for DSR and ₹8,000 per acre for alternative crops** under the *Mera Pani-Meri Virasat* scheme.
  - **Punjab:** Mobile applications such as **i-Khet and Cooperative Machinery Tracker** help farmers access crop residue management machinery.
  - **Chhattisgarh: Gauthans** facilitate the collection and conversion of agricultural residues and cattle dung into organic manure, promoting waste utilisation and rural livelihoods.
- **Improved Satellite Monitoring:** Integrating **polar-orbiting satellites (MODIS and VIIRS) with geostationary satellite observations** can improve detection of stubble-burning incidents beyond limited satellite overpass times and enable more accurate assessment of burned areas.

What Strategies Can India Adopt to Curb Stubble Burning?

- **Policy and Legal Reforms:** A **Parliamentary Committee** recommended establishing a **minimum remunerative price for paddy straw** that covers collection, transportation and other associated costs, making residue utilisation economically viable.
  - Adopt a **balanced approach combining incentives, awareness and proportionate enforcement** , rather than relying primarily on penalties against farmers.
- **Crop Diversification and Procurement Reforms:** Reform **MSP-backed procurement incentives** to encourage farmers to shift from water-intensive paddy cultivation towards alternative crops such as pulses, oilseeds and millets.
  - Promote **short-duration paddy varieties and Direct Seeding of Rice (DSR)** to ease the narrow harvesting-to-sowing window and reduce pressure to burn residues.
- **Infrastructure and Logistics:** Implement **real-time crop mapping, harvest forecasting and decentralised biomass collection systems** , supported by temporary storage facilities, local aggregators and reliable transportation networks.
  - Strengthen **Custom Hiring Centres (CHCs)** to ensure timely and affordable access to Crop Residue Management (CRM) machinery during peak harvesting periods.
- **Innovative Farming Technologies:** Promote **Pusa Decomposer** , a microbial formulation that accelerates stubble decomposition within approximately **20–25 days** , and **Happy Seeders and Super Seeders** for wheat sowing without burning paddy residues.
  - Encourage **in-situ residue management** through mulching and soil incorporation, alongside **ex-situ utilisation** for cattle fodder (where suitable), organic manure, paper, packaging materials, biofuels and power generation.
- **Economic Incentives and Market Development:** Establish **assured markets and remunerative benchmark prices for crop residues** , ensuring that farmers recover collection, labour and transportation costs.
  - Expand **performance-based incentives** for residue management and crop diversification, drawing lessons from Haryana's incentive-based approach.
- **Community-Based Post-Harvest Management:** Promote **MGNREGA -like labour-support mechanisms** , farmer cooperatives and village-level collection centres for residue gathering, composting and transportation.
  - Encourage **Panchayati Raj Institutions (PRIs)** to coordinate residue management and reward villages demonstrating sustained reductions in burning.
- **Improved Monitoring and Accountability:** Integrate **polar-orbiting satellites (MODIS and VIIRS) with geostationary satellite observations** to detect fires beyond fixed satellite overpass times.
  - Use **burned-area assessments, ground verification and transparent reporting** , rather than relying solely on satellite-detected fire counts, to evaluate policy effectiveness.

Conclusion

Curbing stubble burning requires a **holistic approach combining crop diversification, affordable residue management technologies, assured economic incentives and effective policy enforcement** . Strengthening farmer participation, biomass markets and satellite-based monitoring can transform crop residue from an environmental burden into an economic resource, ensuring **sustainable agriculture, improved air quality and long-term environmental security** .

Drishti Mains Question:

Q .Stubble burning in northwestern India reflects deeper structural challenges in agricultural policy rather than merely farmers' non-compliance with environmental regulations. Discuss. (150 words)

Frequently Asked Questions (FAQs)

1. What is stubble burning?
Stubble burning is the practice of setting fire to crop residue left in fields after harvesting.

2. Why is stubble burning common in Punjab and Haryana?
The paddy-wheat cropping cycle, assured procurement and the short interval between harvesting and wheat sowing encourage the practice.

3. What is in-situ crop residue management?
It involves managing crop residue within the field using machinery such as Happy Seeders and Super Seeders.

4. What is ex-situ crop residue management?
It involves collecting and using crop residue outside the field, including for bioethanol and compressed biogas production.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/H8dWT9JA_Kg]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Consider the following agricultural practices: (2012)

1. Contour bunding
2. Relay cropping
3. Zero tillage

In the context of global climate change, which of the above helps/help in carbon sequestration/storage in the soil?

(a) 1 and 2 only

(b) 3 only

(c) 1, 2 and 3

(d) None of them

Ans: (b)

Mains

Q. What are the major factors responsible for making the rice-wheat system a success? In spite of this success, how has this system become bane in India? **(2020)**

Q. Mumbai, Delhi and Kolkata are the three mega cities of the country but the air pollution is much more serious problem in Delhi as compared to the other two. Why is this so? **(2015)**

Indian Coast Guard Inducts Indigenous ALH MK-III Helicopter

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Source: AIR

Recently, the **Indian Coast Guard** inducted the **indigenous Advanced Light Helicopter (ALH) MK-III,** designated **CG 875** , under **Coast Guard Region (West).**

- CG 875 is the **first of five allotted ALH MK-III helicopters** to be positioned with 831 Squadron (CG) at **Ratnagiri, Maharashtra.**

ALH MK-III

- **About:** Advanced Light Helicopter (ALH) MK-III is an **indigenous multi-role helicopter** manufactured by **Hindustan Aeronautics Limited (HAL).**
- **Advanced Equipment:** It features advanced radar, electro-optical sensors, Shakti engines, a full glass cockpit, a high-intensity searchlight, advanced communication systems, an **Automatic Identification System (AIS)** and a search-and-rescue homing system.
- **Operational Role** : It supports **maritime reconnaissance, surveillance and search-and-rescue operations** over extended ranges, operating from ships during both day and night.
- **Multi-role Capability:** The helicopter can switch from an offensive platform equipped with a heavy machine gun to a medical evacuation platform fitted with a medical intensive care unit for transporting critically ill patients.
- **Strategic Significance** : Its induction strengthens the Indian Coast Guard’s maritime surveillance, threat detection, coastal security and emergency response capabilities, including the protection of India’s **Exclusive Economic Zone (EEZ).**

Read more: **Advanced Light Helicopter (ALH) MK-III Squadron**

IMD Launches Interactive Cyclone Warning and Marine Weather Platform

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Source: DD

The **India Meteorological Department (IMD)** launched an interactive **Cyclone Warning and Marine Weather Services Decision Support System(DSS)** to strengthen cyclone preparedness and support timely decision-making.

- **About DSS:** It is a **WebGIS-based platform** providing interactive maps and visualisations of cyclone-related hazards, including cyclone tracks, intensity forecasts, wind distribution, rainfall warnings, storm surge information and rainfall forecasts for river catchments.
- **Background:** The launch coincided with the anniversaries of the landfalls of **Very Severe Cyclonic Storms Phailin (2013)** and Hudhud (2014), whose successful forecasting marked important milestones in India's cyclone early-warning capabilities.
- **Objective:** It provides **location-specific hazard information** to disaster management authorities, government agencies and the public to improve preparedness and reduce losses.
- **Accessibility** : The platform is accessible through IMD and Regional Specialised Meteorological Centre((RSMC) websites and IMD’s Decision Support System.
- **Institutional Coordination:** IMD's **biannual Pre-Cyclone Exercise Meeting** , conducted since 2008, brings together disaster management authorities, defence forces, coastal states, fisheries departments and scientific organisations **to review cyclone preparedness.**
- **Significance:** The platform enables location-specific early warnings, supports evacuation planning and **disaster risk reduction, and improves coordination** among forecasting agencies, disaster management authorities and other stakeholders.

Read more: India’s Cyclone Preparedness

NASA’s Perseverance Rover Sets a New Driving Record on Mars

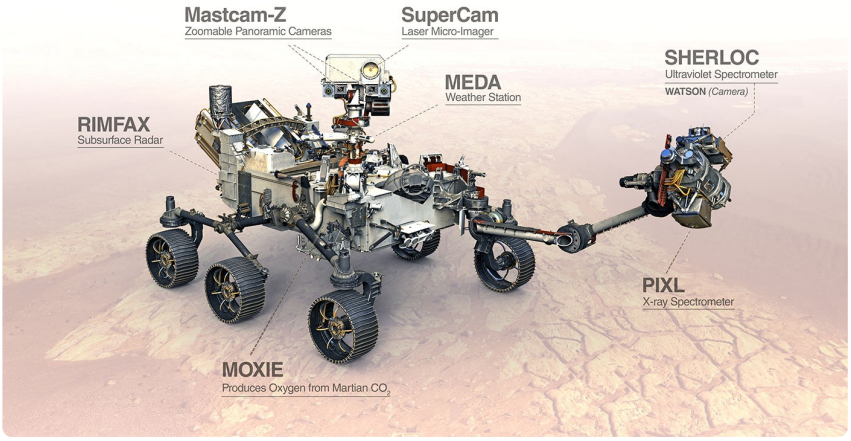
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Source: TP

NASA’s Perseverance rover has travelled 45.22 km on Mars, surpassing the previous off-Earth

driving record of **45.16 km held by NASA's Opportunity rover,** demonstrating the capabilities of autonomous navigation in planetary exploration.

- **About Perseverance:** It is a **car-sized, six-wheeled robotic rover** developed by NASA to investigate **Mars** ' geology, ancient habitability and potential evidence of past microbial life. It primarily explores **Jezero Crater** , a region believed to have contained an **ancient lake** and river delta billions of years ago.
- **Objective** : To investigate evidence of ancient microbial life on Mars, study its geology and evolution, collect rock and regolith samples for potential return to Earth, and identify environments that may have supported life.
- **Important Technologies:** Perseverance uses **AutoNav f** or autonomous navigation, **the Vision Compute Element** for rapid image processing, and **a plutonium-238-powered Multi-Mission Radioisotope Thermoelectric Generator (MMRTG) for electricity generation** .
- **Findings:** NASA has reported a **potential biosignature** in ancient rocks associated with a former river channel feeding Jezero Crater but this is not confirmation that life existed on Mars.
- **Extended Mission:** NASA has extended the mission to explore the crater rim until approximately the end of 2028, although the rover may remain operational beyond this period.
- **Opportunity Rover:** It was launched alongside its twin **Spirit** , landed on Mars in 2004 to investigate the planet's geology and past conditions suitable for microbial life. It found evidence of **ancient habitable conditions on Mars** and operated for nearly 15 years, exceeding its planned mission duration by about 60 times.



Read more: **Ancient Lake on Mars**

Indian Army Launches Bharat Rice

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Source: TH

The **Indian Army** launched ' **Bharat Rice'** for defence personnel and signed a five-year Memorandum of Understanding (MoU) with the **Food Corporation of India (FCI)** , to strengthen military logistics and operational preparedness.

- **About Bharat Rice:** It is a **superior-quality rice initiative** developed based on feedback from troops, with rice supplied according to prescribed **defence specifications to improve ration quality and logistical efficiency.**
- **Objective** : To improve ration quality, ensure timely replenishment and strengthen military preparedness, especially during emergencies and rapid mobilisation.
- **Supply Mechanism** : Rice will be delivered **directly** to Army supply depots, including those in remote areas of Ladakh and the Northeast, reducing the need for depots to collect supplies from local FCI facilities.
- **Role of NAFED:** The **National Agricultural Cooperative Marketing Federation of India (NAFED)** will supply the rice during the financial year 2026–27.
- **Packaging and Quality Control:** Rice will undergo quality checks and be supplied in 10-kg and 30-kg bags. Ergonomic packaging will facilitate handling and transportation in difficult terrain.
- **MoU Highlights:** The agreement gives the Army access to FCI's warehousing infrastructure, modern silos, trained manpower and material-handling equipment during operational contingencies.
  - It will remain valid for **five years,** with an option to extend it by mutual consent.
- **Strategic Significance:** The initiative **strengthens military supply** chains in forward areas, reflecting lessons from **Operation Sindoor** about the importance of resilient logistics during **short-warning or zero-warning** military contingencies.

Read more: **Food Corporation of India**